Séance 4

Jean-Baptiste Guiffard

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button[1] a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```r
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00
```

```
## [1] 4
```

```
## [1] 25
```

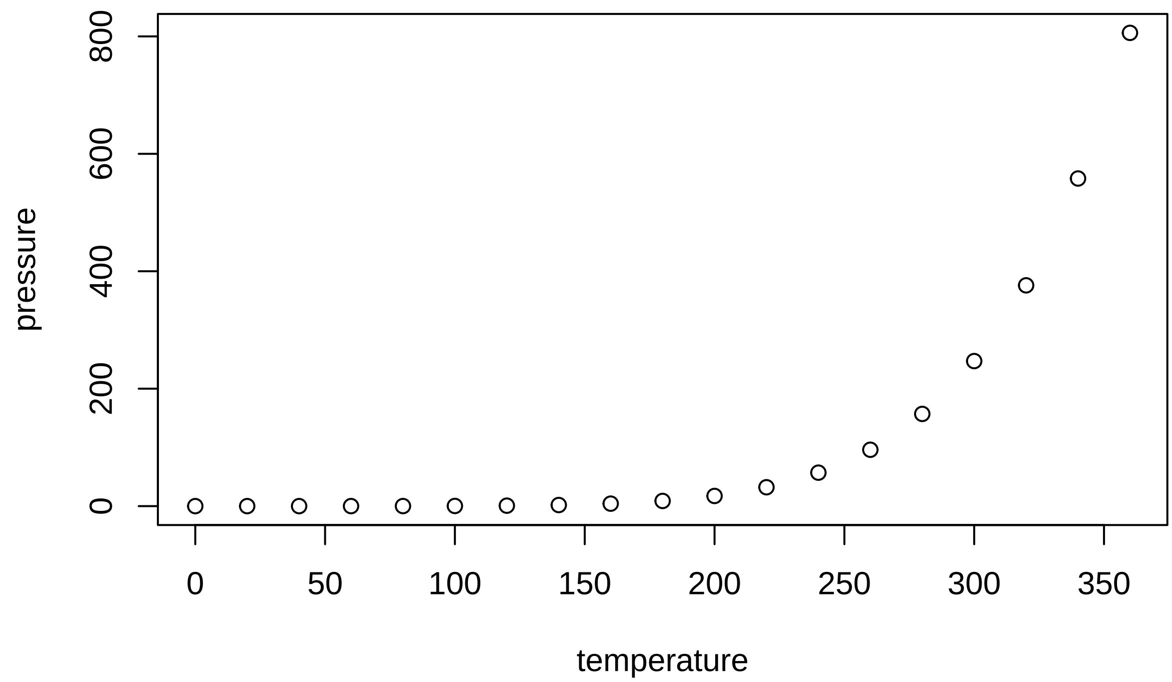
```r
min(cars$speed)
max(cars$speed)
```

La vitesse maximale est de 25 miles per hour, et la vitesse minimale est de 4 miles per hour.

Including Plots

You can also embed plots, for example:

[1] Note de bas de page.



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.Y

$$Y\beta_1 x_1 + \beta_2 x_2 + \epsilon$$

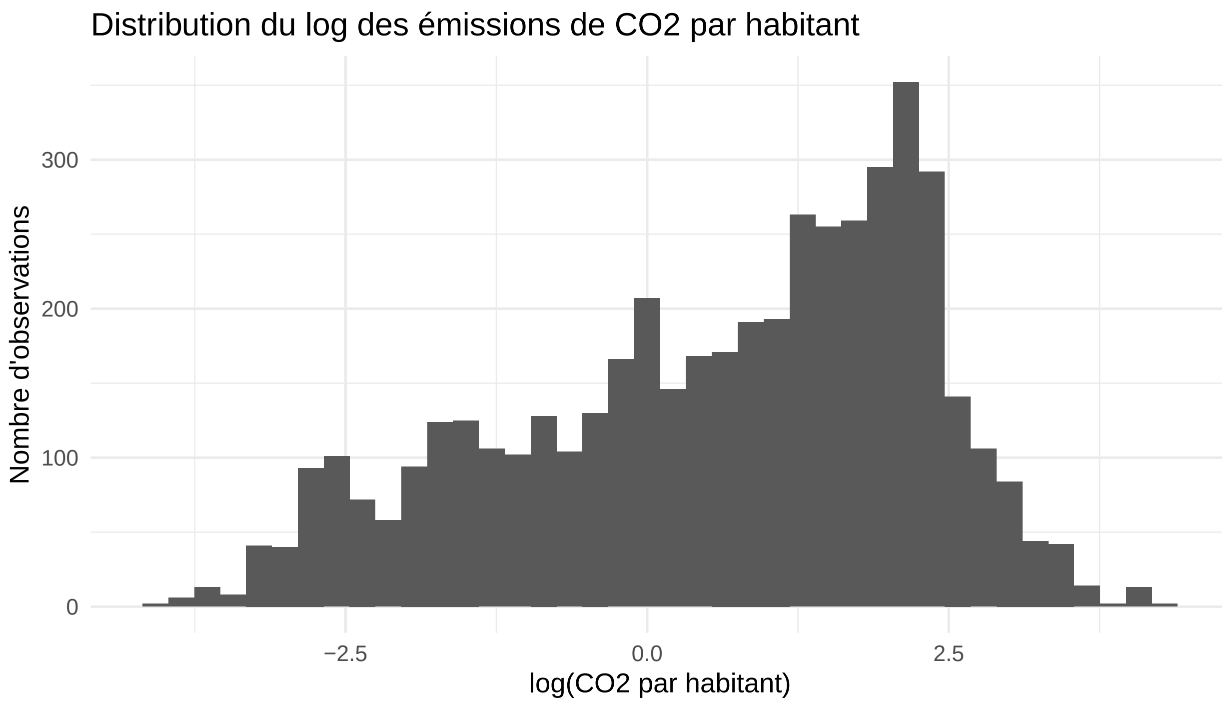
Observations	Pays	CO2 moyen par habitant	PIB moyen par habitant
4753	164	4.9	13745.79

```r
year_stats <- co2_analysis %>% group_by(year) %>% summarise(
mean_co2_pc = mean(co2_per_capita, na.rm = TRUE), median_co2_pc =
median(co2_per_capita, na.rm = TRUE), n_countries = n_distinct(country), .groups = "drop" )

knitr::kable(head(year_stats, 10), digits = 2)
```

year	mean_co2_pc	median_co2_pc	n_countries
1990	4.87	2.53	163
1991	4.61	2.20	163
1992	4.63	2.26	163
1993	4.86	2.14	164
1994	4.83	2.00	164
1995	4.73	2.08	164
1996	4.81	2.10	164
1997	4.85	2.19	164

year	mean_co2_pc	median_co2_pc	n_countries
1998	4.76	2.18	164
1999	4.71	2.14	164



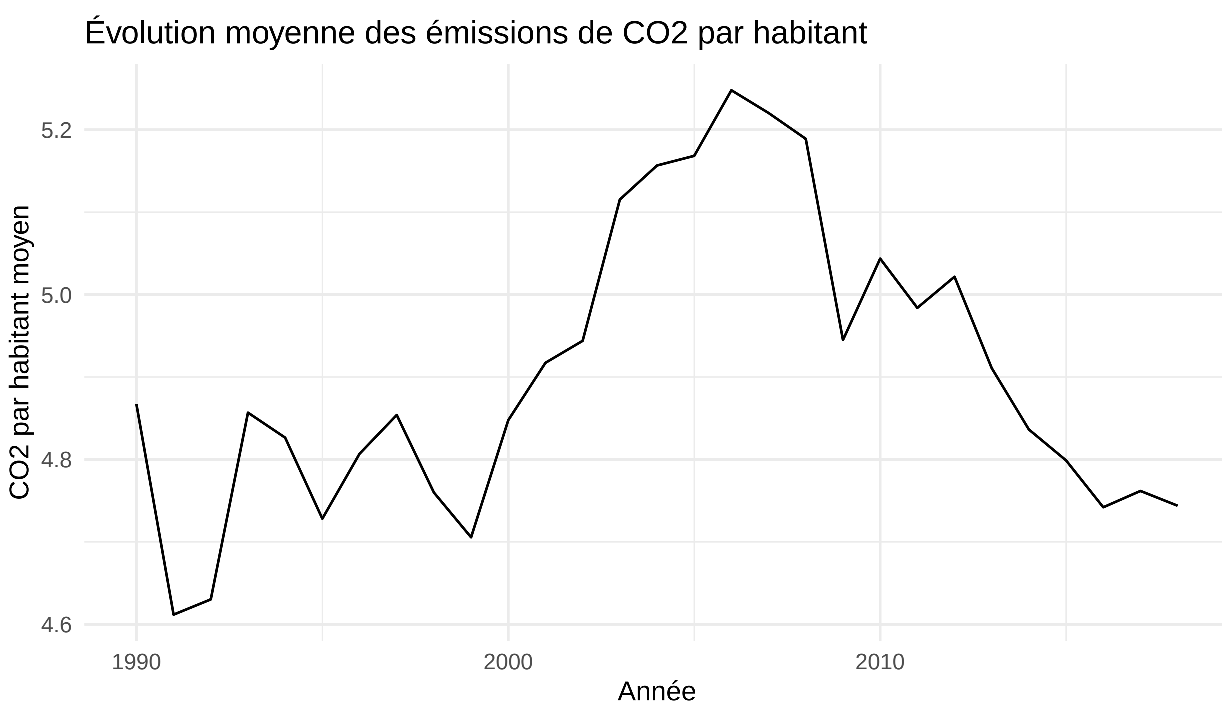


Figure 1: Ma figure

Tableau de régression

```
##
## Call:
## lm(formula = log_co2_pc ~ log_gdp_pc, data = co2_analysis)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.16351 -0.47317 -0.08082  0.37098  2.79019
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -10.678334   0.076918  -138.8   <2e-16 ***
## log_gdp_pc    1.266884   0.008568   147.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7127 on 4751 degrees of freedom
## Multiple R-squared:  0.8215, Adjusted R-squared:  0.8215
## F-statistic: 2.187e+04 on 1 and 4751 DF,  p-value: < 2.2e-16
```

Table 3: Relation entre PIB par habitant et émissions de CO2

	(1)
(Intercept)	-10.678***
	(0.077)
log_gdp_pc	1.267***
	(0.009)
Num.Obs.	4753
R2	0.822
R2 Adj.	0.821
AIC	10 272.7
BIC	10 292.1
Log.Lik.	-5133.350
F	21 865.595
RMSE	0.71

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$